

Assignment 1: Probability & Random Variables — Solutions

Problem 1. We show $P(\limsup_n A_n) = 0$. Define $B_n = \bigcup_{k=n}^{\infty} A_k$; then $\limsup_n A_n = \bigcap_{n=1}^{\infty} B_n$ and $B_1 \supseteq B_2 \supseteq \dots$. By Boole's inequality,

$$P(B_n) \leq \sum_{k=n}^{\infty} P(A_k) =: S_n.$$

Since $\sum_{n=1}^{\infty} P(A_n) < \infty$, the tail $S_n \rightarrow 0$. By continuity of P from above (valid since $P(B_1) \leq \sum P(A_n) < \infty$),

$$P\left(\limsup_n A_n\right) = \lim_{n \rightarrow \infty} P(B_n) \leq \lim_{n \rightarrow \infty} S_n = 0.$$

Since probability is non-negative, $P(\limsup_n A_n) = 0$.

Problem 2. By De Morgan and Boole's inequality:

$$P\left(\bigcap_{i=1}^n A_i\right) = 1 - P\left(\bigcup_{i=1}^n A_i^c\right) \geq 1 - \sum_{i=1}^n (1 - p_i) = \sum_{i=1}^n p_i - (n - 1).$$

Tightness. For $n = 2$ with $p_1 + p_2 > 1$, take $\Omega = [0, 1]$ uniform, $A_1 = [0, p_1)$, $A_2 = [1 - p_2, 1]$. Then $P(A_1 \cap A_2) = p_1 + p_2 - 1$, exactly the lower bound.

Problem 3. The hypothesis $P(A|B) \geq P(A)$ is equivalent (after multiplying by $P(B)$) to

$$P(A \cap B) \geq P(A)P(B). \quad (*)$$

Now compute:

$$P(A^c \cap B^c) - P(A^c)P(B^c) = [1 - P(A) - P(B) + P(A \cap B)] - [(1 - P(A))(1 - P(B))] = P(A \cap B) - P(A)P(B)$$

by (*). Dividing by $P(B^c) > 0$ gives $P(A^c|B^c) \geq P(A^c)$. The statement is **true**.

Problem 4. The probability of exactly k tagged animals in a sample of n from a population of N (with T tagged) is hypergeometric:

$$L(N) = \frac{\binom{T}{k} \binom{N-T}{n-k}}{\binom{N}{n}}.$$

Consider the ratio of consecutive values:

$$\frac{L(N)}{L(N-1)} = \frac{\binom{N-T}{n-k}}{\binom{N-1-T}{n-k}} \cdot \frac{\binom{N-1}{n}}{\binom{N}{n}} = \frac{N-T}{N-T-(n-k)} \cdot \frac{N-n}{N} = \frac{(N-T)(N-n)}{N(N-n-T+k)}.$$

Setting this ≥ 1 and expanding:

$$(N-T)(N-n) \geq N(N-n-T+k) \iff nT \geq kN \iff N \leq \frac{nT}{k}.$$

So $L(N)$ increases for $N \leq nT/k$ and decreases for $N > nT/k$. The MLE is

$$\hat{N} = \left\lfloor \frac{nT}{k} \right\rfloor.$$

Problem 5. The number of permutations of n elements with exactly k fixed points is $\binom{n}{k} D_{n-k}$, where $D_m = m! \sum_{j=0}^m \frac{(-1)^j}{j!}$ is the number of derangements of m elements. Dividing by $n!$:

$$p(n, k) = \frac{\binom{n}{k} D_{n-k}}{n!} = \frac{D_{n-k}}{k! (n-k)!} = \frac{1}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}.$$

As $n \rightarrow \infty$ with k fixed, the sum is the partial sum of $e^{-1} = \sum_{j=0}^{\infty} (-1)^j / j!$. Since the series converges absolutely, the tail satisfies

$$\left| e^{-1} - \sum_{j=0}^{n-k} \frac{(-1)^j}{j!} \right| \leq \sum_{j=n-k+1}^{\infty} \frac{1}{j!} \leq \frac{e}{(n-k+1)!} \rightarrow 0.$$

Therefore $p(n, k) \rightarrow \frac{1}{e \cdot k!}$.

Problem 6. Let $N = r + b$. Conditioning on the first draw (R_1 or B_1):

$P(R_2)$: After R_1 the urn is $(r, b+c)$; after B_1 it is $(r+c, b)$. Both have total $N+c$.

$$P(R_2) = \frac{r}{N+c} \cdot \frac{r}{N} + \frac{r+c}{N+c} \cdot \frac{b}{N} = \frac{r^2 + b(r+c)}{N(N+c)} = \frac{rN + bc}{N(N+c)}.$$

$P(R_2) = P(R_1) = r/N$ iff $rN + bc = r(N+c)$ iff $bc = rc$ iff $b = r$.

$P(R_3)$: Conditioning on all four length-2 histories and tracking urn states before draw 3:

History	Urn before draw 3	Joint prob	$P(R_3 \cdot)$
$R_1 R_2$	$(r, b+2c)$	$\frac{r^2}{N(N+c)}$	$\frac{r}{N+2c}$
$R_1 B_2$	$(r+c, b+c)$	$\frac{rb}{N(N+c)}$	$\frac{r+c}{N+2c}$
$B_1 R_2$	$(r+c, b+c)$	$\frac{b(r+c)}{N(N+c)}$	$\frac{r+c}{N+2c}$
$B_1 B_2$	$(r+2c, b)$	$\frac{b^2}{N(N+c)}$	$\frac{r+2c}{N+2c}$
$P(R_3) = \frac{r^3 + rb(r+c) + b(r+c)^2 + b^2(r+2c)}{N(N+c)(N+2c)}.$			

Since $P(R_1) = r/N$ while $P(R_2) = (rN + bc)/[N(N+c)]$, and these differ whenever $b \neq r$, $P(R_n)$ is **not time-invariant** in general. (Unlike the standard Polya urn where same-color replacement preserves the marginal.)

Problem 7. Let Par_X denote the event that prisoner X is pardoned, each with prior $1/3$. Let W_B = warden names B. With strategy parameter $q = P(W_B|\text{Par}_A) \in [0, 1]$:

$$P(W_B|\text{Par}_B) = 0, \quad P(W_B|\text{Par}_C) = 1.$$

By total probability: $P(W_B) = q/3 + 0 + 1/3 = (q+1)/3$. Bayes gives:

$$P(\text{Par}_A|W_B) = \frac{q/3}{(q+1)/3} = \frac{q}{q+1}, \quad P(\text{Par}_C|W_B) = \frac{1}{q+1}.$$

Under the natural assumption $q = 1/2$ (warden randomises uniformly when A is pardoned):

$$P(\text{Par}_A|W_B) = \frac{1}{3}, \quad P(\text{Par}_C|W_B) = \frac{2}{3}.$$

Prisoner A's probability remains $1/3$; he is **wrong**. The information transfers entirely to C.

Problem 8. Alice wins on her k -th turn iff all $2(k-1)$ preceding tosses are Tails, and toss $2k-1$ is Heads:

$$P(A_k) = (1-p)^{2(k-1)} p.$$

Summing the mutually exclusive events:

$$P(\text{Alice wins}) = \sum_{k=1}^{\infty} p(1-p)^{2(k-1)} = \frac{p}{1-(1-p)^2} = \frac{p}{p(2-p)} = \frac{1}{2-p}.$$

For fairness we need $1/(2-p) = 1/2$, giving $p = 0$, which is outside $(0,1)$. Since $1/(2-p) > 1/2$ for all $p \in (0,1)$, **no fair bias exists**.

Problem 9. The condition $P(A \cap B \cap C) = P(A)P(B)P(C)$ **does not** imply mutual independence.

Counterexample. Let $\Omega = \{1, 2, 3, 4\}$ with uniform measure $P(\{i\}) = 1/4$. Set $A = \{1, 2\}$, $B = \{1, 2\}$, $C = \{1, 3\}$. Then:

$$P(A \cap B \cap C) = P(\{1\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \checkmark$$

but

$$P(A \cap B) = \frac{1}{2} \neq P(A)P(B) = \frac{1}{4}.$$

So A and B are not independent despite the triple product condition holding.

Problem 10. Let $\Delta(x) = F(x) - F(x^-)$ be the jump of F at x (which equals $P(X = x)$). For any $\varepsilon > 0$, the set $D_\varepsilon = \{x : \Delta(x) > \varepsilon\}$ must be finite, since otherwise $\sum_{x \in D_\varepsilon} \Delta(x) > \sum_{x \in D_\varepsilon} \varepsilon = \infty$, contradicting $\sum_x \Delta(x) \leq F(+\infty) - F(-\infty) = 1$.

The full set of discontinuities is

$$D = \{x : \Delta(x) > 0\} = \bigcup_{n=1}^{\infty} D_{1/n},$$

a countable union of finite sets, hence **at most countable**.

Problem 11. Enumerate $\mathbb{Q} \cap [0, 1] = \{q_1, q_2, \dots\}$. Define the discrete distribution $P(X = q_n) = 2^{-n}$; since $\sum_{n=1}^{\infty} 2^{-n} = 1$, this is valid. Set

$$F(x) = \sum_{n=1}^{\infty} \frac{1}{2^n} \mathbf{1}_{[q_n \leq x]}.$$

This is non-decreasing, right-continuous (each q_n has positive mass only there), and satisfies $F(-\infty) = 0$, $F(+\infty) = 1$.

Strict monotonicity on $(0, 1)$: For any $a < b$ in $(0, 1)$, density of $\mathbb{Q}$ guarantees some $q_n \in (a, b)$, so $F(b) - F(a) \geq 2^{-n} > 0$.

Problem 12. For $r \in [0, 1]$, the event $\{X \leq r\}$ corresponds to the disk of radius r :

$$F_X(r) = \frac{\pi r^2}{\pi \cdot 1^2} = r^2.$$

Differentiating: $f_X(r) = 2r$ for $r \in [0, 1]$, zero elsewhere. Verification: $\int_0^1 2r \, dr = 1$.

Problem 13. For $y \in [0, 1]$:

$$F_Y(y) = P(X^2 \leq y) = P(|X| \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} \, dx = \sqrt{y}.$$

Differentiating: $f_Y(y) = \frac{1}{2\sqrt{y}}$ for $y \in (0, 1)$, zero elsewhere.

Problem 15. Since F is continuous, $P(X_2 = x) = F(x) - F(x^-) = 0$ for every $x \in \mathbb{R}$. By independence and the law of total probability:

$$P(X_1 = X_2) = \int_{-\infty}^{\infty} P(X_2 = x) \, dF(x) = \int_{-\infty}^{\infty} 0 \, dF(x) = 0.$$

More rigorously: let $D_n = \{|X_1 - X_2| < 1/n\}$, so $\{X_1 = X_2\} = \bigcap_n D_n$. Then

$$P(D_n) = \int_{-\infty}^{\infty} [F(x + \frac{1}{n}) - F(x - \frac{1}{n})] \, dF(x) \xrightarrow{n \rightarrow \infty} 0$$

by dominated convergence, since the integrand $\rightarrow F(x^+) - F(x^-) = 0$ pointwise.

Problem 16. The CDF of X is:

$$F_X(x) = \begin{cases} 0 & x < 0, \\ p & x = 0, \\ p + (1-p)G(x) & x > 0. \end{cases}$$

F_X has a jump of size p at 0 and is strictly increasing for $x > 0$ (since G is).

The q -th quantile $\xi_q = \inf\{x : F_X(x) \geq q\}$:

- If $q \leq p$: $F_X(0) = p \geq q$ and $F_X(x) < q$ for all $x < 0$, so $\xi_q = 0$.
- If $q > p$: we need $p + (1-p)G(x) \geq q$, i.e. $G(x) \geq \frac{q-p}{1-p}$, giving

$$\xi_q = G^{-1}\left(\frac{q-p}{1-p}\right).$$

$$\boxed{\xi_q = \begin{cases} 0 & q \leq p, \\ G^{-1}\left(\frac{q-p}{1-p}\right) & q > p. \end{cases}}$$